Dual-tower MOF adsorption predictor

Actual checkpoint architecture | 14,536,691 trainable parameters | T = 298 K

CO_2: 0.01 / 0.05 / 0.1 / 0.5 / 2.5 bar
CH_4: 0.05 / 0.5 / 0.9 / 2.5 / 4.5 bar

10 uptake values (mmol g^{-1})

Positive, monotonic transform — separately for each gas
q = expm1(min(cumsum(softplus(r)), 6))

Shared MLP head: 256 → 256 → 1
SiLU

First 10 query-token outputs

Fusion Transformer encoder × 4
8 heads · d = 256 · FFN = 1024
Pre-LN blocks + final LayerNorm

10 learned fusion queries
+ condition tokens
10 × 256

Concatenate [10 queries | 10 structure | 10 descriptor]
30 × 256 + token-type embeddings

Structure auxiliary
MLP → monotonic q

Descriptor auxiliary
MLP → monotonic q

Condition adapter × 2
Transformer + LN → 10 × 256

Condition adapter × 2
Transformer + LN → 10 × 256

Gas + pressure
5 pressure features
6 gas properties
Gas ID + condition ID
10 × 256

Structure tower

Invariant attention pooling
8 learned queries → 8 × 256

Equivariant graph attention × 6
8 heads · scalar: 256
vector: 48 × 3

Edge context
64 RBF
+ 24 relation dims
+ soft bond score
↓
MLP → 256

Unit directions

Atom embedding + node MLP
Z embedding + [13 features; 13 masks]
scalar $h \in \mathbb{R}^{256}$; vector v = 0

CIF → periodic atomic graph
6 Å cutoff · max. 32 neighbors/atom
Z, node features, edge vectors

Descriptor tower

Final LayerNorm
31 × 256 tokens

Transformer encoder × 4
8 heads · FFN = 1024 · Pre-LN

Feature-wise tokenization
$t_j = x_j w_j + b_j + e_j$
30 feature tokens + 1 [CLS]

30 geometry + chemistry descriptors
Median imputation + RobustScaler

Dashed branches: auxiliary supervision. Modality dropout (p = 0.05) is applied before fusion during training.